

5 mm Square Surface Mount Miniature Trimmers Single-Turn Cermet Sealed



FEATURES

- 0.25 W at 70 °C
- For through hole version see T53Y series
- Wide ohmic range (10 Ω to 1 M Ω)
- Small size for optimum packaging density
- Tests according to CECC 41000 or IEC 60393-1
- Construction: fully sealed
- Professional grade
- Material categorization: for definitions of compliance please see www.vishay.com/doc299912


RoHS
COMPLIANT

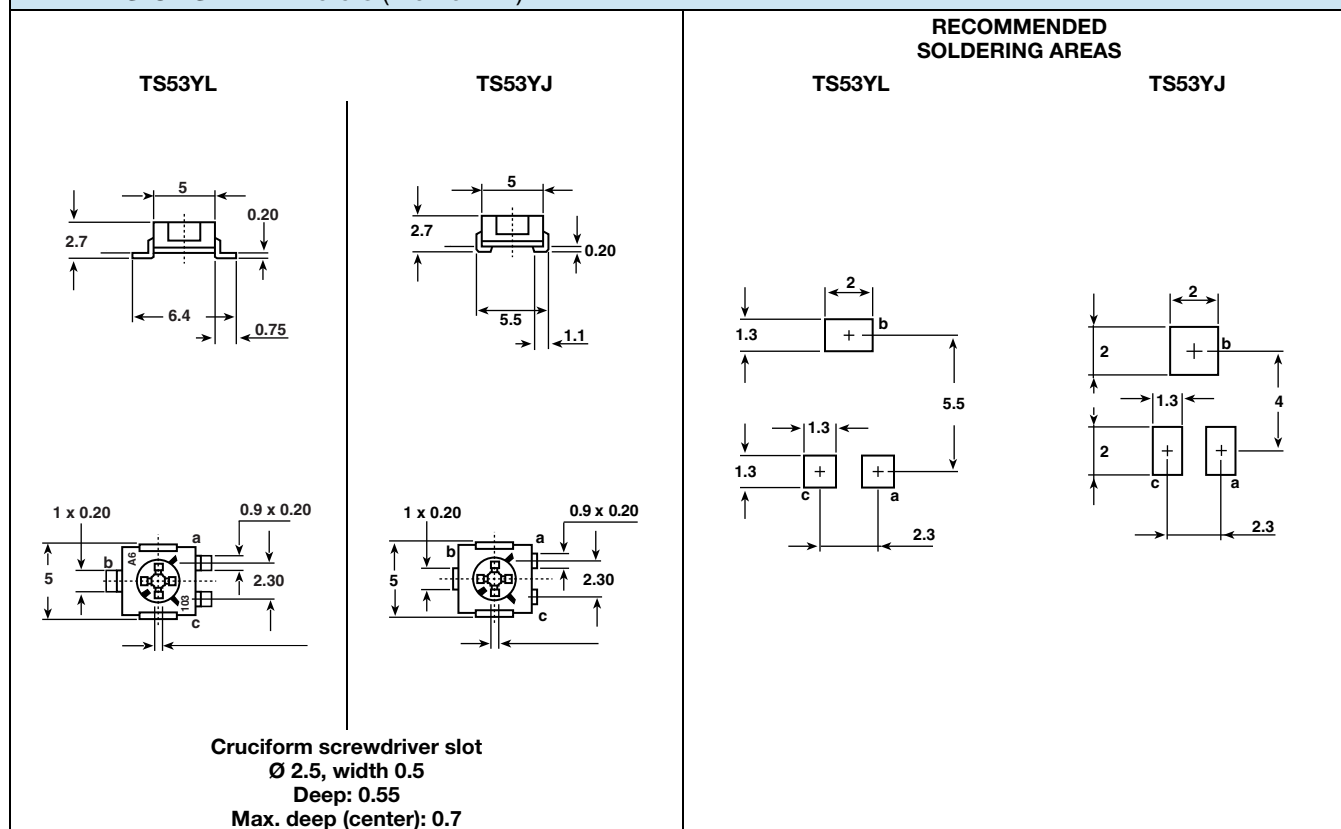
LINKS TO ADDITIONAL RESOURCES

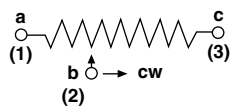
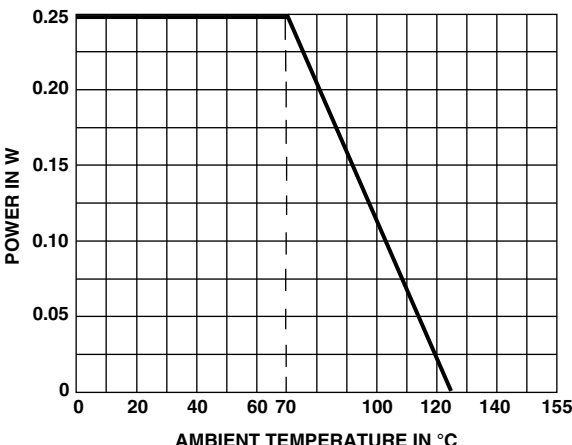


The TS53 trimming potentiometer has been designed for surface mount applications and offers volumetric efficiency (5 mm x 5 mm x 2.7 mm) with high performance and stability.

The TS53 design is suitable for both manual or automatic operation, and can withstand wave, and reflow soldering techniques.

DIMENSIONS in millimeters (± 0.25 mm)



ELECTRICAL SPECIFICATIONS	
Resistive element	Cermet
Electrical travel	$220^\circ \pm 15^\circ$
Resistance range	10Ω to $1 \text{ M}\Omega$
Standard series	1 - 2 - 5
Tolerance standard	$\pm 20 \%$
Circuit diagram	
Power rating	Linear 0.25 W at $+70^\circ \text{C}$ 
Temperature coefficient	$\pm 100 \text{ ppm}/^\circ \text{C}$
Limiting element voltage (linear law)	200 V
Contact resistance variation (typical)	1 % or 3Ω
End resistance (typical)	0.1 % or 3Ω
Dielectric strength (RMS)	1000 V
Insulation resistance	$1 \text{ G}\Omega$

MECHANICAL SPECIFICATIONS	
Mechanical travel	$270^\circ \pm 10^\circ$
Operating torque (max. Ncm)	1.5
End stop torque (max. Ncm)	3.5
Unit weight (max. g)	0.15
Terminals	Pure Sn (e3)

ENVIRONMENTAL SPECIFICATIONS	
Temperature range	-55°C to $+125^\circ \text{C}$
Climatic category	55 / 125 / 56
Sealing	Sealed container IP67
MSL level	4

SOLDERING RECOMMENDATIONS	
Recommended reflow profile 2, see Application Note www.vishay.com/doc?52029	
Caution	
Reflow soldering must be done within 72 h while stored under a max. temperature of 30°C , 60 % RH after opening the dry pack envelope.	

RECOMMENDED METHOD OF STORAGE

Dry box storage is recommended as soon as the hermetic bag has been opened to prevent moisture absorption. The following conditions should be observed, if dry boxes are not available:

- Storage temperature 10 °C to 30 °C
- Storage humidity ≤ 60 % RH max.

After more than 72 h under these conditions, moisture content will be too high for reflow soldering.

In case of moisture absorption, the devices will recover to the former condition by drying under the following condition:

192 h at 40 °C + 5 °C/- 0 °C and < 5 % RH (dry air/nitrogen) or

96 h at 60 °C + 5 °C and < 5 % RH for all device containers (not suitable for reel) or

24 h at 125 °C + 5 °C (not suitable for reel)

PERFORMANCES

TESTS	CONDITIONS	TYPICAL VALUES AND DRIFTS		
		$\Delta R_T/R_T$ (%)	$\Delta R_{1-2}/R_{1-2}$ (%)	OTHER
Electrical endurance	1000 h at rated power 90'/30' - ambient temp. 70 °C	± 2 %	± 3 %	Contact resistance variation: $\Delta R < 1 \% R_n$
Climatic sequence	Phase A dry heat 125 °C Phase B damp heat Phase C cold -55 °C Phase D damp heat 5 cycles	± 2 %	± 3 %	
Damp heat steady state	Temperature 40 °C - RH 93 % 56 days	± 2 %	± 3 %	Dielectric strength: 1000 V _{RMS} Insulation resistance: > 10 ⁴ MΩ
Charge of temperature	-55 °C to +125 °C - 5 cycles	± 1 %		$\Delta V_{1-2}/V_{1-3} \leq \pm 2 \%$
Mechanical endurance	100 cycles - rated power	± (3 % + 5 Ω)		
Shock	50 g - 11 ms 3 successive shocks in 3 directions	± 1 %		$\Delta V_{1-2}/V_{1-3} \leq \pm 1 \%$
Vibration	10 Hz to 55 Hz 0.75 mm or 10 g - 6 h	± 1 %		$\Delta V_{1-2}/V_{1-3} \leq \pm 1 \%$

Note

- Nothing stated herein shall be construed as a guarantee of quality or durability.

STANDARD RESISTANCE ELEMENT DATA

STANDARD RESISTANCE VALUES	LINEAR LAW			TYPICAL TCR -55 °C +125 °C
	MAX. POWER AT 70 °C	MAX. WORKING VOLTAGE	MAX. CURRENT THROUGH ELEMENT	
Ω	W	V	mA	ppm/°C
10	0.25	1.58	158	± 100
20	0.25	2.24	112	
50	0.25	3.54	71	
100	0.25	5.00	50	
200	0.25	7.07	35	
500	0.25	11.2	22	
1K	0.25	15.8	16	
2K	0.25	22.4	11	
5K	0.25	35.4	7	
10K	0.25	50.0	5	
20K	0.25	70.7	3.5	
50K	0.25	112	2.2	
100K	0.25	158	1.6	
200K	0.20	200	1.0	
500K	0.08	200	0.4	
1M	0.04	200	0.2	

MARKING

Vishay trademark, ohmic value, manufacturing date

The ohmic value is indicated by a 3 figure code, the first two are significant figures, the third one is the multiplier.

Example: 100 = 10 Ω

101 = 100 Ω

102 = 1000 Ω

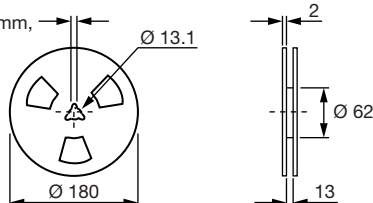
503 = 50 000 Ω

PACKAGING

On tape and reel of 500 pieces, code R10 (TR500) and 2000 pieces, code R20 (TR2000)

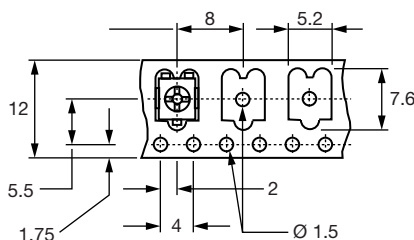
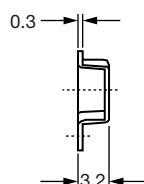
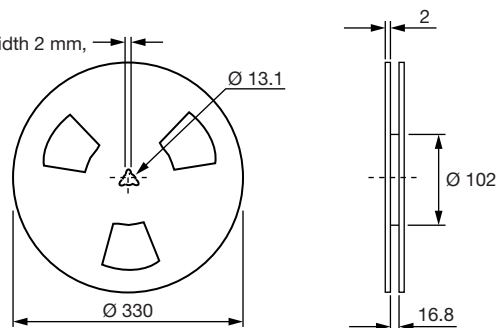
R10

3 slots - width 2 mm,
each 120°



R20

3 slots - width 2 mm,
each 120°



Cover tape panel strength specifications EIA 481 A and CEI 60286-3.

DRYPACK

Devices are packed in moisture barrier bags to prevent the products from moisture absorption during transportation and storage. Each bag contains a desiccant.

ORDERING INFORMATION (part number)

T	S	5	3	Y	L	5	0	4	M	R	1	0				
MODEL		STYLE		OHMIC VALUE		TOLERANCE		PACKAGING		SPECIAL NUMBER						
TS53		YL YJ		From 10 Ω to 1 M Ω 504 = 500 k Ω		M = 20 %		R10 = reel 500 pieces On request R20 = reel 2000 pieces		(If applicable) Given by Vishay for custom design						



DESCRIPTION (for information only)

TS53	YL	500K	20 %		TR	e3
MODEL	STYLE	VALUE	TOLERANCE	SPECIAL	PACKAGING	LEAD (Pb)-FREE

RELATED DOCUMENTS

APPLICATION NOTES

Potentiometers and Trimmers	www.vishay.com/doc?51001
Guidelines for Vishay Sfernice Resistive and Inductive Components	www.vishay.com/doc?52029
Selector guide	www.vishay.com/doc?49286

ACCESSORIES

Screwdrivers (to order separately)	www.vishay.com/doc?57015
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